Lithium Diffusion in a Spherical Particle: Effect of Flux Shape, Frequency and Diffusivity

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1 Problem

Lithium diffusion inside a spherical particle is studied using a PDE model. The concentration inside the particle varies with radius and time, and lithium enters or leaves only through the surface. The outputs looked at are the surface concentration C_s, the average concentration C_{avg}, and the concentration profile $C(r)$ inside the particle. Different flux inputs and different diffusivity values are applied to the same particle to see how they change the response. Six experiments are reported. Experiments 1 to 4 change one thing at a time (24 PDE simulations). Experiments 5 and 6 build two-equation models for C_s and C_{avg}, by the Galerkin method and by SINDy, and test them against further PDE simulations.

2 Model

The model is Fick's second law in spherical coordinates, written in dimensionless form. The scalings used are

$$\tau = \frac{D_0 t}{R^2}, \qquad r = \frac{\tilde{r}}{R}, \qquad C = \frac{c}{c_0}, \qquad f(C) = \frac{D(c)}{D_0}, \qquad \delta = \frac{jR}{D_0 c_0}, \tag{1}$$

so τ is scaled time, r runs from 0 at the centre to 1 at the surface, C is concentration as a fraction of full, $f(C)$ is the diffusivity relative to the reference value D_0, and δ is the scaled surface flux. The governing equation is

$$\frac{\partial C}{\partial \tau} = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2 f(C)\frac{\partial C}{\partial r}\right], \tag{2}$$

with boundary and initial conditions

$$f(C)\left.\frac{\partial C}{\partial r}\right|_{r=1} = -\delta(\tau), \tag{3}$$

$$\left.\frac{\partial C}{\partial r}\right|_{r=0} = 0, \tag{4}$$

$$C(r,0) = 1. \tag{5}$$

The two reported quantities are

$$C_s(\tau) = C(1,\tau), \qquad C_{\text{avg}}(\tau) = 3\int_0^1 C(r,\tau)\, r^2\, dr. \tag{6}$$

Integrating Equation (2) over the particle and using the boundary conditions gives

$$\frac{dC_{\text{avg}}}{d\tau} = -3\,\delta(\tau), \tag{7}$$

which ties the average concentration directly to the applied flux.

2.1 Parameters

The particle radius is $R = 2 \times 10^{-6}$ m, the maximum concentration is $c_0 = 51554$ mol m^{-3}, and the reference diffusivity is $D_0 = 2 \times 10^{-14}$ m^2 s^{-1}. These give $R^2/D_0 = 200$ s, so one unit of τ is 200 seconds. The particle starts full and uniform, $C_0 = 1$, and each run of Experiments 1 to 4 covers $0 \le \tau \le 10$, that is 2000 seconds. The reference surface flux is $j_0 = 4.3 \times 10^{-6}$ mol m^{-2} s^{-1}, which gives $\delta_0 = j_0 R/(D_0 c_0) = 0.008341$. A C-rate multiplier of 3 is used, so the flux amplitude in every deterministic profile is

$$A = 3\,\delta_0 = 0.025022. \tag{8}$$

The equations are solved in MATLAB using `pdepe` in spherical mode on a grid of 1000 radial points and 2000 time points, with relative and absolute tolerances of 10^{-9} and a maximum step of 10^{-2} in τ. These settings are the same for all 24 runs of Experiments 1 to 4. Experiments 5 and 6 use their own run lengths and time steps, given there.

2.2 Diffusivity

Unless stated otherwise the diffusivity is the fitted NMC811 curve used in the solver:

$$f(C) = p_4 + (p_1 - p_4)\exp\left[-\exp\left(-p_2(C - p_3)\right)\right], \tag{9}$$

with $p_1 = 0.1008$, $p_2 = -8.2105$, $p_3 = 0.6986$ and $p_4 = 0.0009$. The NMC811 diffusivity is concentration dependent and is used as the default material model. It is low near $C = 1$, about 0.0009, and rises to about 0.1 at low C. Experiment 3 replaces it with a constant f.

2.3 Flux profiles

Six deterministic flux shapes are used:

$$\texttt{const}: \quad \delta(\tau) = A, \tag{10}$$

$$\texttt{stepUp}: \quad \delta(\tau) = A\left[1 + \tfrac{1}{2}\tanh 20(\tau - 5)\right], \tag{11}$$

$$\texttt{linear}: \quad \delta(\tau) = A\,\frac{\tau}{10}, \tag{12}$$

$$\texttt{linearDown}: \quad \delta(\tau) = A\left(1 - \frac{\tau}{10}\right), \tag{13}$$

$$\texttt{sin}: \quad \delta(\tau) = A\sin(\omega\tau), \tag{14}$$

$$\texttt{cos}: \quad \delta(\tau) = A\cos(\omega\tau). \tag{15}$$

The frequency is $\omega = 20$ everywhere except in Experiment 2, where it is swept. Since `sin` and `cos` are negative for half of each cycle, they push lithium back into the particle, so C can rise above 1 in those runs. The model puts no upper bound on C.

3 Experiment 1: flux shape

- The six flux profiles were applied to the same particle with the same diffusivity model.
- Only the shape of $\delta(\tau)$ was changed; $\omega = 20$ for `sin` and `cos`.

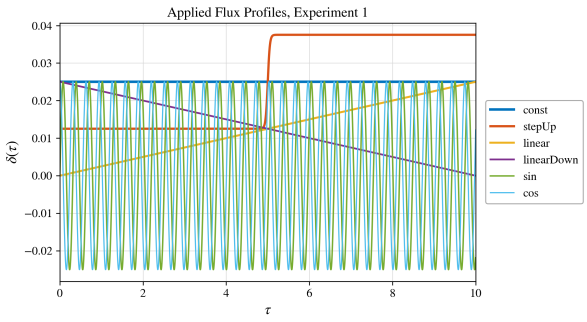


Figure 1: The six applied flux profiles $\delta(\tau)$.

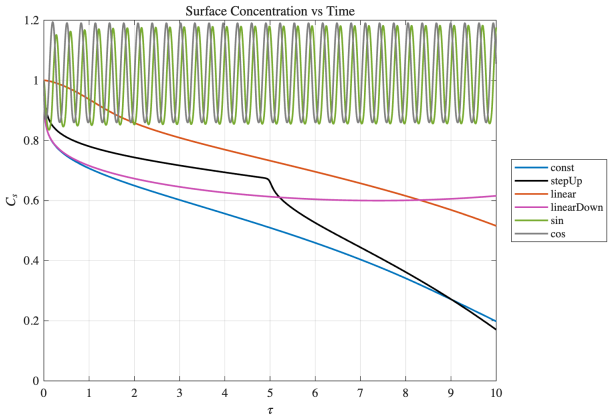


Figure 2: Surface concentration C_s against τ.

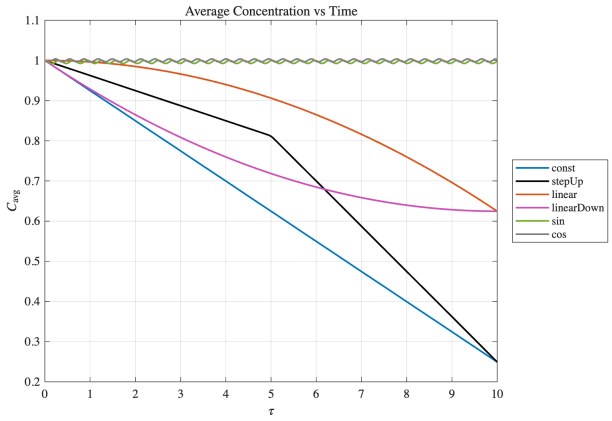


Figure 3: Average concentration C_{avg} against τ.

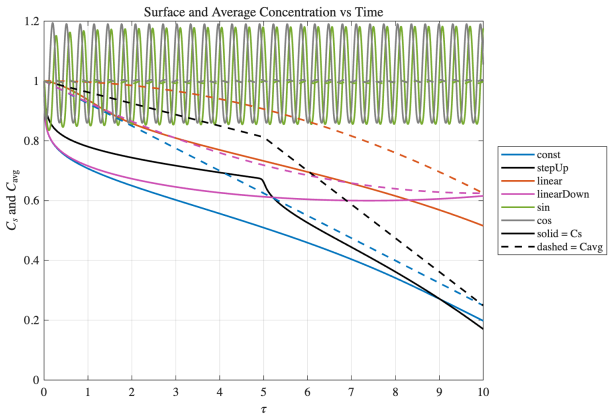


Figure 4: C_s and C_{avg} against τ.

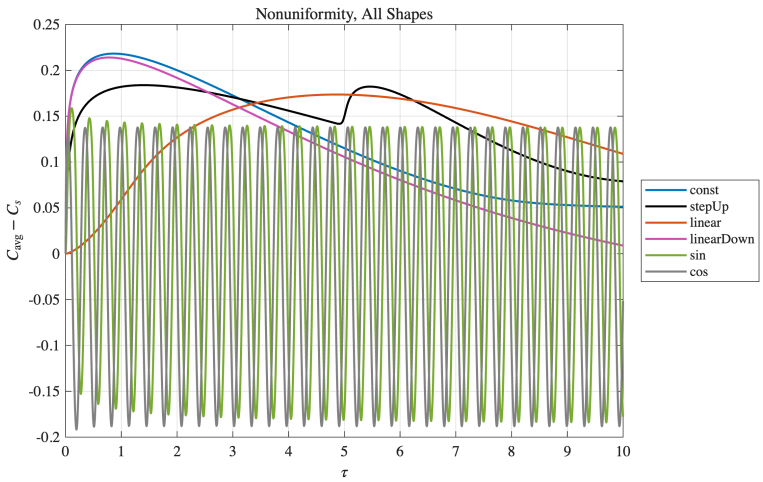


Figure 5: $C_{\text{avg}} - C_s$ against τ.

- C_s changes with the flux shape, while C_{avg} follows the total charge applied.

- `const` and `stepUp`: same charge 0.7507, same final $C_{\text{avg}} = 0.2493$, but C_s minima of 0.1981 and 0.1704.
- `linear` and `linearDown`: same charge 0.3753, same final $C_{\text{avg}} = 0.6247$, but C_s minima of 0.5158 and 0.5999.
- This follows from Equation (7): C_{avg} depends only on the integral of δ.
- Profiles that load the surface harder or earlier give a larger gap. `linearDown` peaks at $\tau = 0$, where the diffusivity is lowest, and reaches a maximum gap of 0.2140 against 0.1736 for `const`.

3.1 Inside the particle

- $r = 0$ is the centre and $r = 1$ is the surface.
- A flat curve means the particle is uniform; a drop near $r = 1$ means the surface is emptied faster than diffusion refills it.

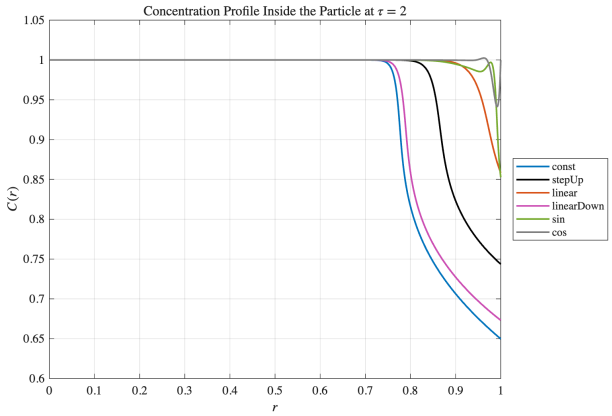


Figure 6: Concentration profile at $\tau = 2$.

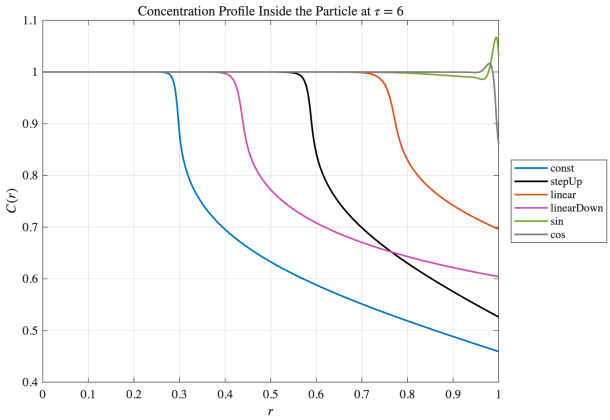


Figure 7: Concentration profile at $\tau = 6$.

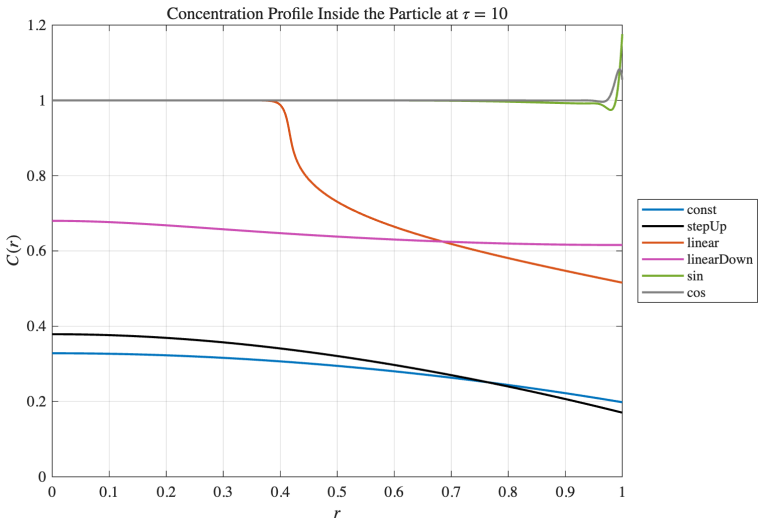


Figure 8: Concentration profile at $\tau = 10$.

- The steady and ramped profiles change the concentration across the whole particle.
- `sin` and `cos` disturb only a thin layer near the surface and leave the interior close to its initial value.

4 Experiment 2: frequency of the oscillating flux

- The flux was fixed at $\delta(\tau) = A\sin(\omega\tau)$ and only ω was changed, taking 2, 5, 10, 20, 40 and 80.
- The material model and all other solver settings were kept fixed.
- The period is $T_\tau = 2\pi/\omega$, which gives 628.3, 251.3, 125.7, 62.8, 31.4 and 15.7 seconds.
- The time plots use the window $5 \leq \tau \leq 6$ (200 seconds), because at $\omega = 80$ the full run holds more than 120 oscillations. Line thickness decreases as ω increases. The full solution is kept in the saved data.

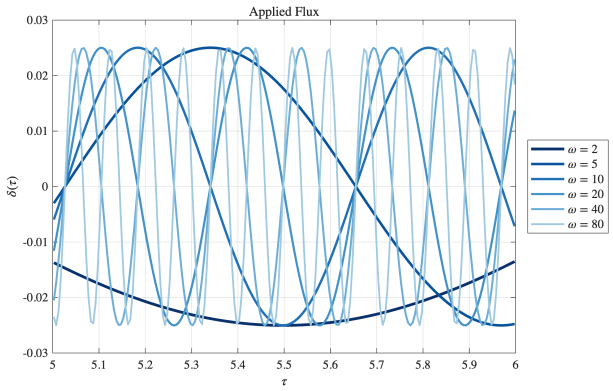


Figure 9: Applied flux $\delta(\tau)$ for the six frequencies.

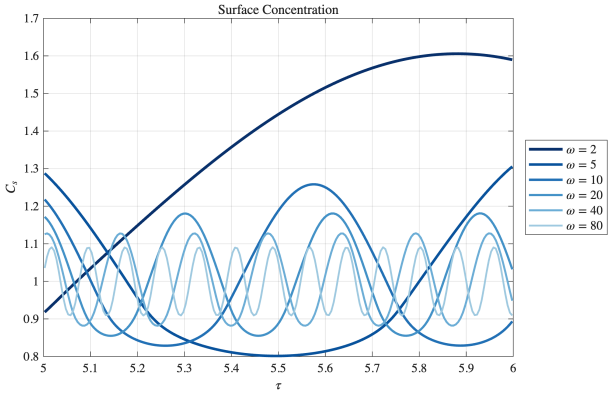


Figure 10: Surface concentration C_s against τ.

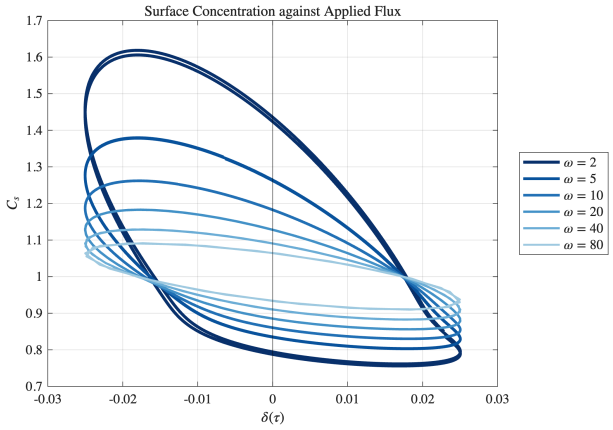


Figure 11: Surface concentration C_s against applied flux δ.

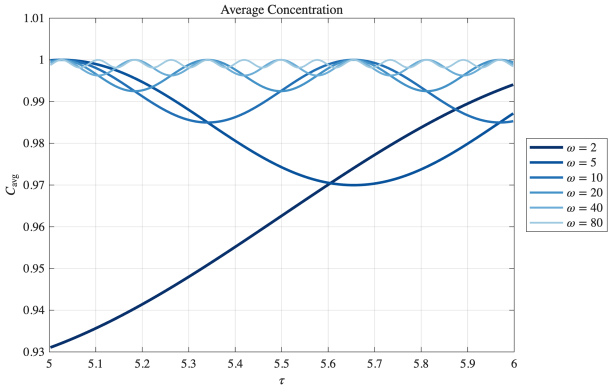


Figure 12: Average concentration C_{avg} against τ.

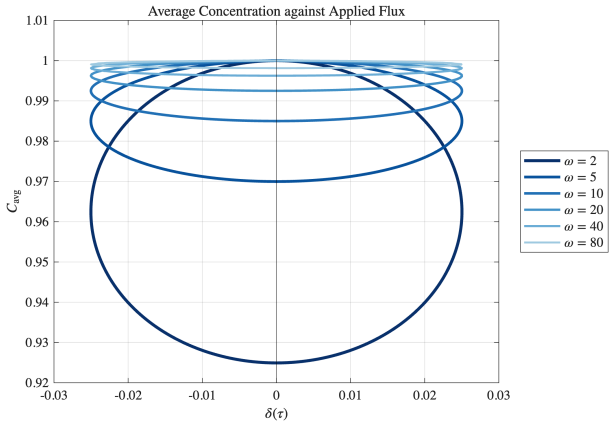


Figure 13: Average concentration C_{avg} against applied flux δ.

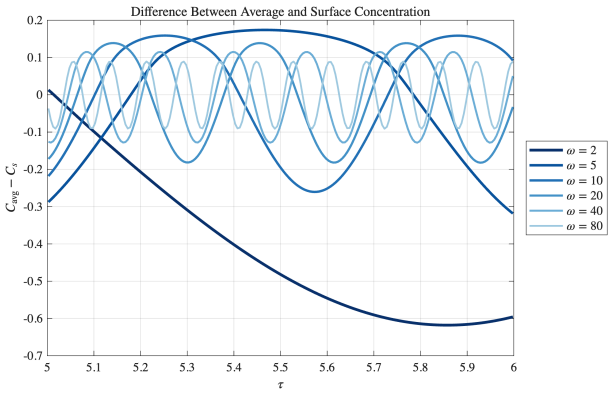


Figure 14: $C_{\text{avg}} - C_s$ against τ.

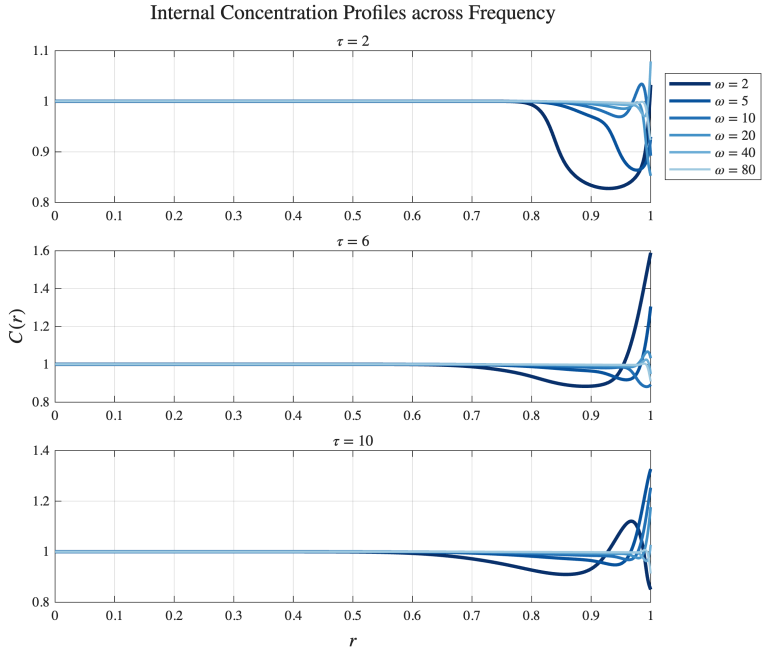


Figure 15: Concentration profiles at $\tau = 2$, 6 and 10.

- Increasing ω reduces the visible oscillation in C_s: it spans 0.729 to 1.618 at $\omega = 2$ and 0.889 to 1.092 at $\omega = 80$, with the intermediate cases in order.
- C_{avg} changes very little. Its minimum rises from 0.925 to 0.998, because a faster oscillation reverses sign before much charge has accumulated.
- C_s against δ is a loop, not a single line, so C_s is not set by the instantaneous flux alone. The loops shrink as ω increases.
- The disturbed region moves closer to the surface: the profile is curved from about $r = 0.55$ at $\omega = 2$, and flat until about $r = 0.97$ at $\omega = 80$.
- The maximum gap $C_{\text{avg}} - C_s$ falls from 0.215 to 0.110.

5 Experiment 3: constant diffusivity

- Equation (9) was replaced by a constant, so $f(C) = f$ at every concentration, taking 0.01, 0.1, 1, 10 and 100.
- The flux was fixed at $\delta(\tau) = A\sin(20\tau)$ and all other settings were kept fixed.
- D_0 was not changed, so all five runs share the same time axis. The same window $5 \leq \tau \leq 6$ is used.

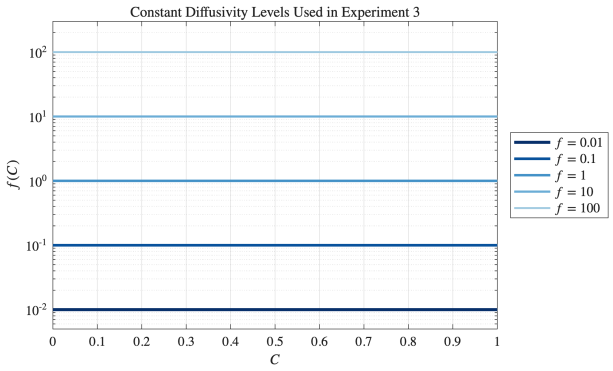


Figure 16: The five constant diffusivity levels (log vertical axis).

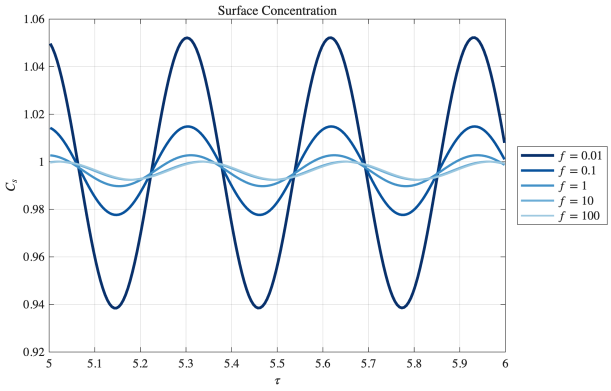


Figure 17: Surface concentration C_s against τ.

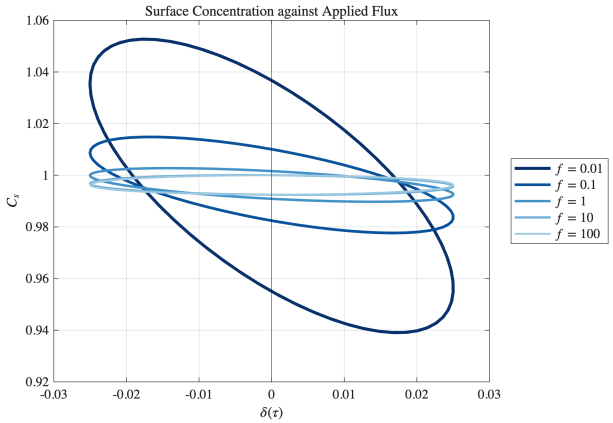


Figure 18: Surface concentration C_s against applied flux δ.

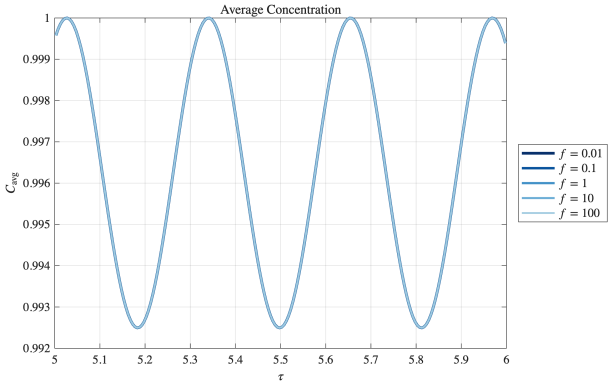


Figure 19: Average concentration C_{avg} against τ.

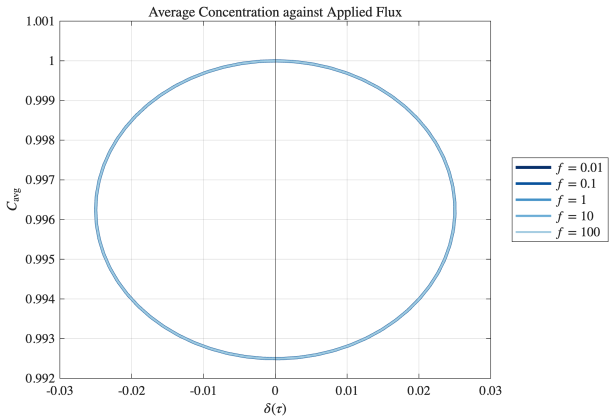


Figure 20: Average concentration C_{avg} against applied flux δ.

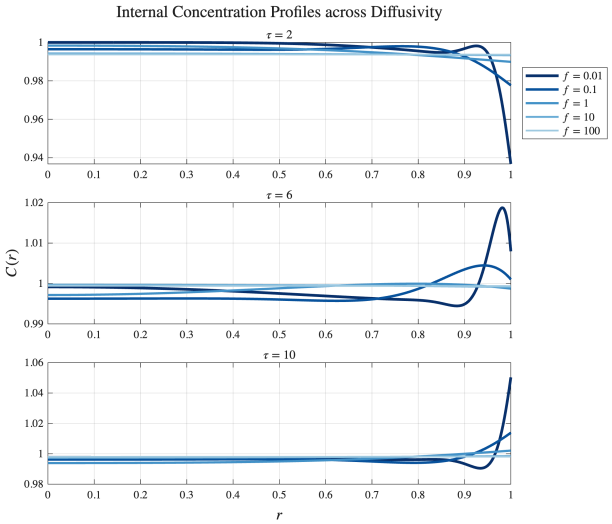


Figure 21: Concentration profiles at $\tau = 2$, 6 and 10.

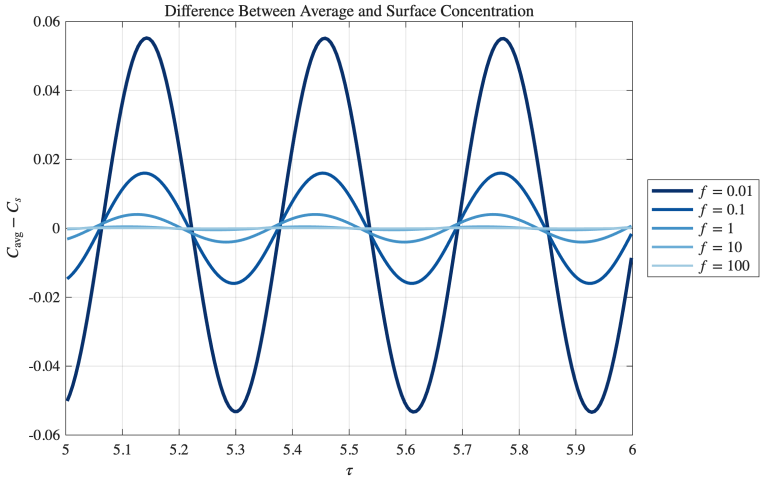


Figure 22: $C_{\text{avg}} - C_s$ against τ.

- At low diffusivity the profile is strongly non-uniform near the surface. As f increases the gradient becomes smaller and the profile approaches a flat line.
- At $f = 0.01$ the profile is flat through the interior and turns sharply in the last few percent of the radius, reaching 1.05 at the surface at $\tau = 10$ while the centre sits near 0.996. At $f = 100$ it is flat across the whole radius.
- The C_s range narrows with f: 0.9228–1.0527, then 0.9740–1.0148, 0.9897–1.0028, 0.9924–1.0001 and 0.99249–1.00000.
- The maximum gap $C_{\text{avg}} - C_s$ falls from 0.0710 to 0.0198, 0.00418, 0.000499 and 0.000051.
- The C_{avg} histories are essentially the same in all five runs, spanning 0.99249 to 1.00000 with a final value of 0.99808, because the applied flux is the same and Equation (7) contains no diffusivity.

6 Experiment 4: randomly generated flux

- `trueRandom` generates a reproducible piecewise-constant flux from a fixed seed. It divides the run into 17 segments of random length and gives each segment a random amplitude.
- Changing the seed gives a different excitation while the generation method and all solver settings stay unchanged. This is useful for generating varied input–output data for later modelling.
- Five seeds were used: 30303, the original seed in the project and the reference case, then 101, 202, 303 and 404.
- Only the flux and the surface concentration are plotted.
- The amplitude convention differs from the deterministic profiles: an internal constant of 0.01 is used instead of $A = 0.025022$, and the mean is subtracted from the amplitude sequence. The net charge removed is -0.024 to $+0.018$, against 0.751 for a steady discharge at $\delta = A$, so these are excitation signals rather than discharges and should not be compared directly with Experiment 1.

6.1 Individual seeds

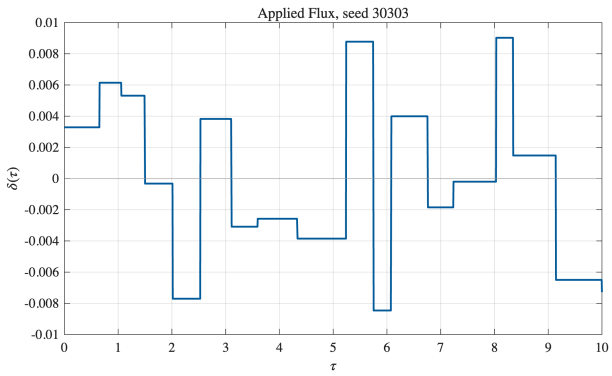


Figure 23: Seed 30303 (reference). Applied flux $\delta(\tau)$.

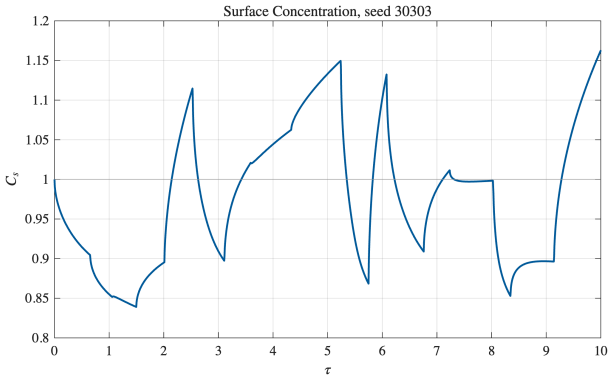


Figure 24: Seed 30303. Surface concentration C_s.

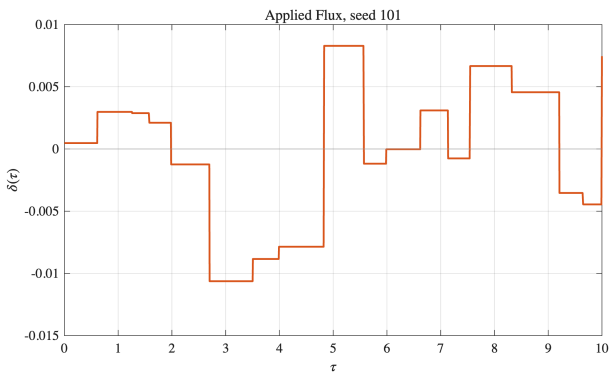


Figure 25: Seed 101. Applied flux $\delta(\tau)$.

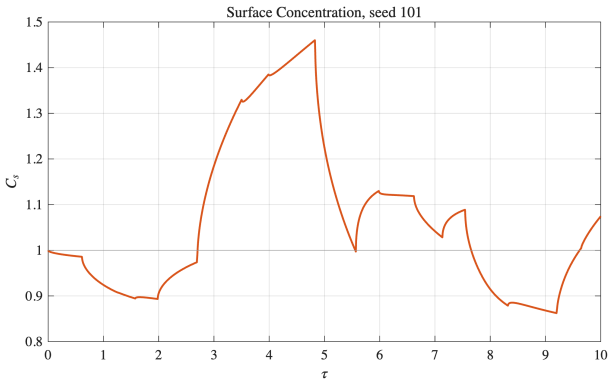


Figure 26: Seed 101. Surface concentration C_s.

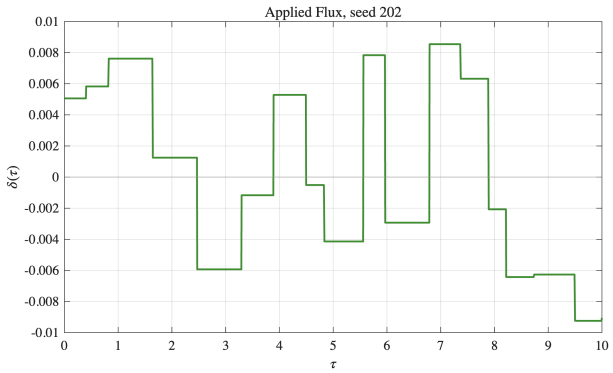


Figure 27: Seed 202. Applied flux $\delta(\tau)$.

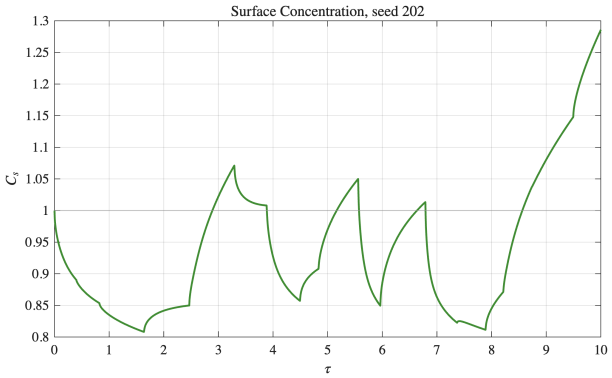


Figure 28: Seed 202. Surface concentration C_s.

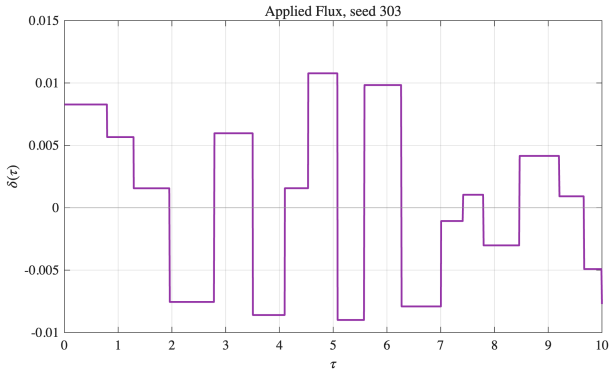


Figure 29: Seed 303. Applied flux $\delta(\tau)$.

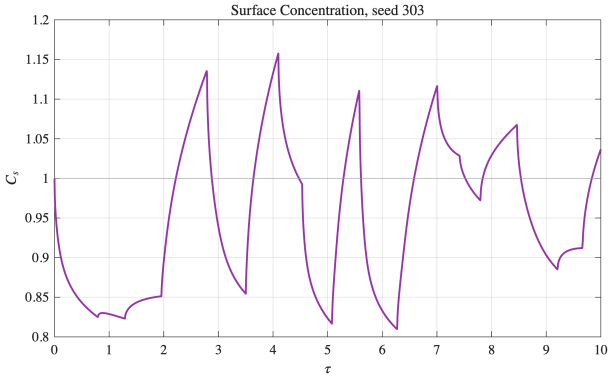


Figure 30: Seed 303. Surface concentration C_s.

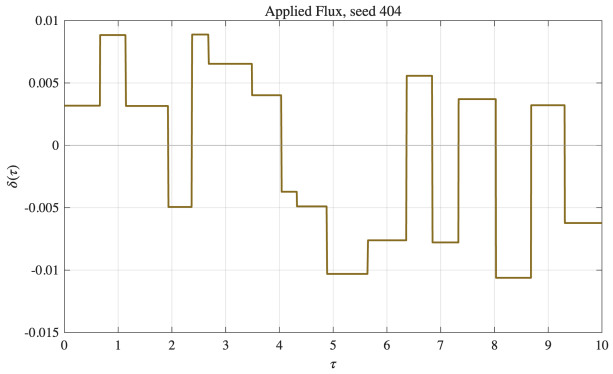


Figure 31: Seed 404. Applied flux $\delta(\tau)$.

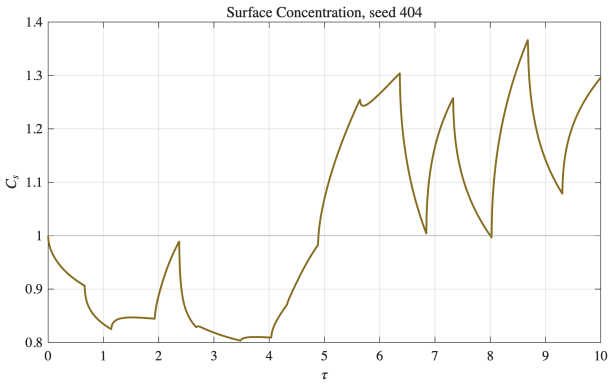


Figure 32: Seed 404. Surface concentration C_s.

6.2 All seeds together

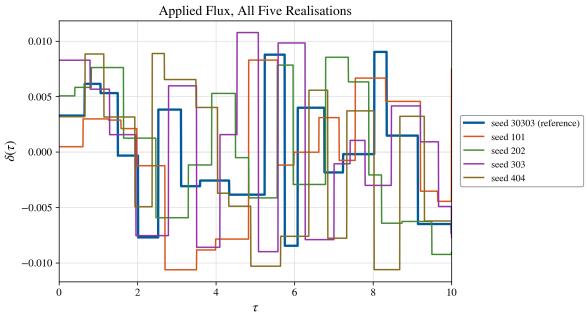


Figure 33: Applied flux $\delta(\tau)$ for all five seeds.

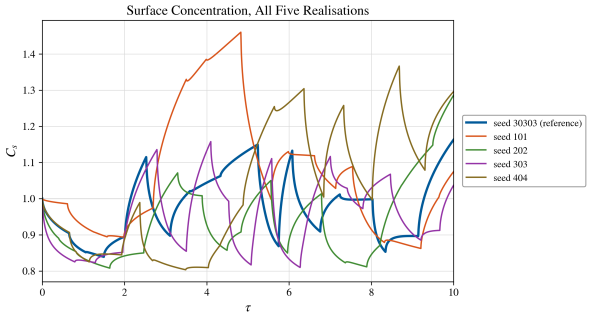


Figure 34: Surface concentration C_s for all five seeds.

- The five fluxes differ from each other, with segment edges at different times and different sign patterns.
- The surface responses are similar in character: irregular excursions above and below $C_s = 1$ with no steady drift.
- Across the five seeds C_s spans 0.803 to 1.460 and C_{avg} spans 0.949 to 1.049.
- The minimum of C_s varies only between 0.803 and 0.862, while the maximum varies more widely, between 1.158 and 1.460.
- Five seeds only show that the behaviour is similar in character, not anything about its distribution. The per-seed values are saved in `Results/tables/trueRandom_summary.csv`.

7 Experiment 5: two-equation model by the Galerkin method

- The aim is a model of two ordinary differential equations, one for C_{avg} and one for C_s, derived from the PDE by the Galerkin method, and then tested against the PDE.
- The diffusivity is constant, $f = 1$, throughout this experiment and Experiment 6.

7.1 Starting point: the Galerkin formulas

- The concentration is approximated by N fixed radial shapes $P_k(r)$, each multiplied by a coefficient that depends only on time:

$$C(r,\tau) \approx \sum_{k=1}^{N} \alpha_k(\tau)\, P_k(r). \tag{16}$$

- The shapes are orthogonal with the spherical weight r^2, and the first shape is $P_1 = 1$:

$$\int_0^1 r^2\, P_i(r)\, P_j(r)\, dr = 0 \quad (i \neq j), \qquad P_1(r) = 1. \tag{17}$$

- The Galerkin projection of the PDE onto these shapes gives the following two results, in dimensionless form. They are the starting point of this experiment.

$$\frac{dC_s}{d\tau} = -\delta(\tau) \sum_{i=1}^{N} \frac{P_i(1)^2}{\int_0^1 r^2 P_i(r)^2\, dr} - \sum_{i=1}^{N} \frac{P_i(1) \int_0^1 r^2 P_i'(r)\, f\left(\sum_{k=1}^N \alpha_k P_k(r)\right) \left(\sum_{k=1}^N \alpha_k P_k'(r)\right) dr}{\int_0^1 r^2 P_i(r)^2\, dr}, \tag{18}$$

$$\frac{dC_{\text{avg}}}{d\tau} = \frac{d\alpha_1}{d\tau} = \frac{-\delta(\tau)}{\int_0^1 r^2\, dr} = -3\,\delta(\tau). \tag{19}$$

- $f(\cdot)$ is the diffusivity evaluated at the approximate concentration. With constant f, it is taken out of the integral in Equation (18).
- Equation (19) is the exact mass balance, Equation (7).
- Equation (18) gives $dC_s/d\tau$ in terms of the coefficients α_k. To obtain a model in C_s and C_{avg}, the shapes are chosen and the α_k are written in terms of C_s and C_{avg}.

7.2 Choice of shapes, $N = 2$

- $P_1 = 1$. The second shape is taken as $P_2 = r^2 + c$, the simplest shape that is even in r (so $\partial C/\partial r = 0$ at the centre) and can bend.
- The constant c is fixed by the orthogonality condition, Equation (17), between P_1 and P_2:

$$\int_0^1 r^2 \cdot 1 \cdot (r^2 + c)\, dr = \frac{1}{5} + \frac{c}{3} = 0 \quad \Rightarrow \quad c = -\frac{3}{5}, \qquad P_1(r) = 1, \qquad P_2(r) = r^2 - \frac{3}{5}. \tag{20}$$

- The approximation is $C = \alpha_1 + \alpha_2 \left(r^2 - \frac{3}{5}\right)$, a parabola in r.

7.3 Linking α_1, α_2 to C_{avg} and C_s

Average concentration. Using the definition $C_{\text{avg}} = 3\int_0^1 C(r,\tau)\, r^2\, dr$ and the approximation:

$$C_{\text{avg}} = 3\left[\alpha_1 \int_0^1 r^2\, dr + \alpha_2 \int_0^1 r^2 \left(r^2 - \tfrac{3}{5}\right) dr\right] = 3\left[\frac{\alpha_1}{3} + \alpha_2 \cdot 0\right] = \alpha_1. \tag{21}$$

- The second integral is zero by the orthogonality condition, since $\int_0^1 r^2 P_2\,dr = \int_0^1 r^2 P_1 P_2\,dr = 0$. So the first coefficient is exactly the average concentration, $\alpha_1 = C_{\text{avg}}$, which is why Equation (19) is written as $dC_{\text{avg}}/d\tau$.

Surface concentration. At $r = 1$:

$$C_s = \alpha_1 P_1(1) + \alpha_2 P_2(1) = \alpha_1 + \frac{2}{5}\,\alpha_2. \tag{22}$$

- With $\alpha_1 = C_{\text{avg}}$:

$$\alpha_1 = C_{\text{avg}}, \qquad \alpha_2 = \frac{5}{2}\,(C_s - C_{\text{avg}})\,. \tag{23}$$

7.4 Applying the formula for $dC_s/d\tau$

	$P_i(1)$	$\int_0^1 r^2 P_i^2\,dr$	$P_i'(r)$
$P_1 = 1$	1	1/3	0
$P_2 = r^2 - 3/5$	2/5	4/175	$2r$

Table 1: Values needed in Equation (18).

- $\int_0^1 r^2 P_2^2\,dr = \int_0^1 \left(r^6 - \frac{6}{5}r^4 + \frac{9}{25}r^2\right) dr = \frac{1}{7} - \frac{6}{25} + \frac{3}{25} = \frac{4}{175}$.
- **First sum (flux term):**

$$\frac{P_1(1)^2}{1/3} + \frac{P_2(1)^2}{4/175} = \frac{1}{1/3} + \frac{4/25}{4/175} = 3 + 7 = 10.$$

- **Second sum (diffusion term).** Since $P_1' = 0$, only $i = 2$ contributes, and inside the bracket only $k = 2$ contributes: $\sum_k \alpha_k P_k'(r) = 2r\,\alpha_2$. With constant f:

$$\frac{P_2(1)\,f \int_0^1 r^2\,(2r)(2r\,\alpha_2)\,dr}{4/175} = \frac{\frac{2}{5}\cdot\frac{4}{5}\,f\,\alpha_2}{4/175} = 14\,f\,\alpha_2.$$

- So Equation (18) becomes

$$\frac{dC_s}{d\tau} = -10\,\delta - 14\,f\,\alpha_2.$$

- Replace α_2 using Equation (23): $14f\alpha_2 = 14f \cdot \frac{5}{2}(C_s - C_{\text{avg}}) = 35f(C_s - C_{\text{avg}})$.

7.5 The two-equation Galerkin model

$$\boxed{\frac{dC_{\text{avg}}}{d\tau} = -3\,\delta(\tau), \qquad \frac{dC_s}{d\tau} = -35f\,(C_s - C_{\text{avg}}) - 10\,\delta(\tau), \qquad C_s(0) = C_{\text{avg}}(0) = 1.} \tag{24}$$

- With $f = 1$, written term by term: $dC_s/d\tau = -35\,C_s + 35\,C_{\text{avg}} - 10\,\delta$. These three coefficients are compared with SINDy in Experiment 6.
- The initial values follow from $C(r, 0) = 1$: $\alpha_1 = 1$, $\alpha_2 = 0$.
- Assumptions: constant f; the profile is a combination of 1 and r^2 at every instant; the particle starts uniform.
- Slow-flux check: when δ changes slowly, C_s and C_{avg} move together, so $dC_s/d\tau \approx -3\delta$, which gives $C_s - C_{\text{avg}} = -\delta/(5f)$.
- With $N = 2$ the equation for C_s contains only C_s, C_{avg} and δ. With $N \geq 3$ it would also contain $\alpha_3, \ldots$, which cannot be written with C_s and C_{avg} alone.

7.6 Test against the PDE

- The flux is $\delta(\tau) = A\cos(\omega\tau)$ at $\omega = 0.5$, 1, 2, 3, 5, 10, 20, 30, 50, 100 and 200.
- Each run lasts $\tau_{\max} = \max(10,\ 5 \text{ periods})$ with time step $\min(0.005,\ \text{period}/100)$, on the same 1000-point radial grid.
- The Galerkin model, Equation (24), is solved on the same time points, starting from $C_s = C_{\text{avg}} = 1$.

7.6.1 Error measure

For a predicted signal $\hat{y}$ and a reference y, sampled at the n points $t_1, \ldots, t_n$ of a time *window* W,

$$\text{Rel}_W(\hat{y}, y) = \frac{\text{RMS}_W(\hat{y} - y)}{\max_W y - \min_W y}, \qquad \text{RMS}_W(e) = \sqrt{\frac{1}{n}\sum_{k=1}^{n} e(t_k)^2}, \tag{25}$$

given in % (100×Rel). The denominator is the *band* of the reference: its maximum minus its minimum in the window. The numerator and the denominator use the same window, stated with every result. Here $\hat{y} = C_{s,\text{model}}$ and $y = C_{s,\text{PDE}}$:

$$\text{Rel (\%)} = 100 \times \frac{\text{RMS}\left(C_{s,\text{model}} - C_{s,\text{PDE}}\right)}{\max C_{s,\text{PDE}} - \min C_{s,\text{PDE}}}.$$

ω	band of $C_{s,\text{PDE}}$ (last period)	RMS absolute error (last period)	Rel (%)
0.5	0.3006	3.99×10^{-7}	0.0001
1	0.1507	1.60×10^{-6}	0.0011
2	0.0763	6.37×10^{-6}	0.0084
3	0.0518	1.42×10^{-5}	0.0275
5	0.0329	3.85×10^{-5}	0.1172
10	0.0196	1.38×10^{-4}	0.7031
20	0.0130	3.90×10^{-4}	2.9984
30	0.0104	5.95×10^{-4}	5.7293
50	0.0078	7.95×10^{-4}	10.1721
100	0.0054	8.69×10^{-4}	16.1875
200	0.0037	7.79×10^{-4}	20.9499

Table 2: Band of $C_{s,\text{PDE}}$, absolute and relative error of the Galerkin model, last period.

7.6.2 Windows

- **Whole run:** from $\tau = 0$, where the flux is switched on at full size on a uniform particle, to the end of the run.
- **Last period:** the last period of the run.

7.6.3 Periodicity check

With constant f and a sine or cosine flux, consecutive periods of the PDE solution were compared, and the Galerkin error was computed over the last one and the last two periods:

ω	$\max \lvert C_s(\tau) - C_s(\tau - P)\rvert$ over the last period	Rel %, last 2 periods	Rel %, last period
0.5	1.2×10^{-7}	0.0001	0.0001
3	4.3×10^{-9}	0.0275	0.0275
30	7.5×10^{-8}	5.717	5.729
200	8.9×10^{-10}	20.996	20.950

Table 3: Periodicity of the PDE solution, and the Galerkin error over one and two periods.

The last period is used from here on.

7.6.4 Result

ω	C_s range PDE	C_s range Galerkin	max $\lvert C_s - C_{s,\mathrm{PDE}}\rvert$ whole run	max $\lvert C_s - C_{s,\mathrm{PDE}}\rvert$ last period	Rel (%) whole run	Rel (%) last period
0.5	0.84971–1.15029	0.84971–1.15029	9.50×10^{-4}	5.75×10^{-7}	0.005	0.0001
1	0.92463–1.07537	0.92463–1.07537	9.50×10^{-4}	2.29×10^{-6}	0.015	0.0011
2	0.96186–1.03814	0.96185–1.03815	9.50×10^{-4}	9.02×10^{-6}	0.043	0.0084
3	0.97408–1.02592	0.97407–1.02593	9.50×10^{-4}	2.01×10^{-5}	0.081	0.0275
5	0.98357–1.01643	0.98354–1.01646	9.50×10^{-4}	5.45×10^{-5}	0.172	0.1172
10	0.99022–1.00978	0.99003–1.00997	9.49×10^{-4}	1.95×10^{-4}	0.735	0.7031
20	0.99349–1.00652	0.99299–1.00701	9.40×10^{-4}	5.51×10^{-4}	2.998	2.9984
30	0.99481–1.00524	0.99425–1.00577	9.35×10^{-4}	8.37×10^{-4}	5.672	5.7293
50	0.99609–1.00398	0.99581–1.00427	1.13×10^{-3}	1.13×10^{-3}	10.109	10.1721
100	0.99732–1.00275	0.99763–1.00249	1.26×10^{-3}	1.23×10^{-3}	16.046	16.1875
200	0.99814–1.00191	0.99877–1.00130	1.13×10^{-3}	1.11×10^{-3}	20.752	20.9499

Table 4: Galerkin model against the PDE, cosine flux, $f = 1$.

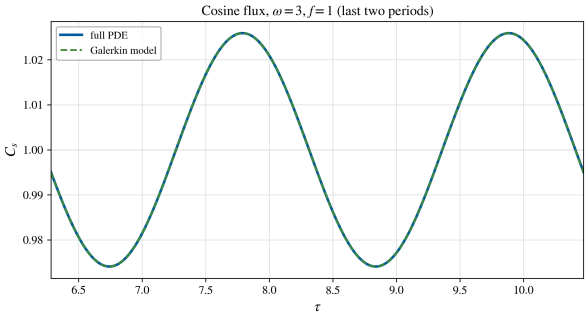


Figure 35: C_s, PDE and Galerkin model, $\omega = 3$.

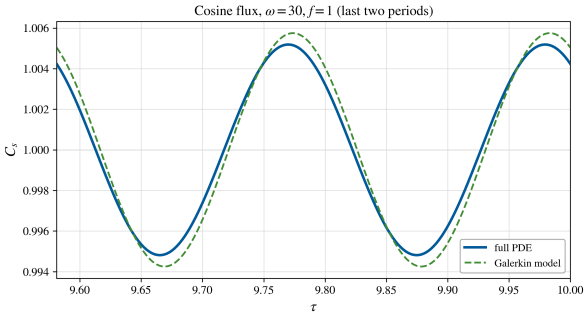


Figure 36: C_s, PDE and Galerkin model, $\omega = 30$.

8 Experiment 6: SINDy

- SINDy (sparse identification of nonlinear dynamics, Brunton, Proctor and Kutz, 2016) finds an equation directly from data. The time derivative of a quantity is written as a combination of candidate terms (the library), and only the terms that are needed are kept.
- The aim is to find the equation for $dC_s/d\tau$ from PDE data and compare its coefficients with the Galerkin model of Experiment 5.
- The SINDy code and the method of choosing λ are first applied to the Lorenz system, whose equations are known.

8.1 Method

- Data are collected at many times and stacked: $\dot{X} = \Theta(X)\,\Xi$, where each column of Θ is one candidate term evaluated on the data, and Ξ holds the unknown coefficients.
- Ξ is found by sequentially thresholded least squares (STLSQ): fit by least squares, set every coefficient smaller than a threshold λ to zero, fit again using only the remaining terms, and repeat (10 times). The threshold is applied to the coefficients directly.

8.1.1 Training, validation and test data

- **Training data**: the data SINDy is fitted on.
- **Validation data**: runs not used in the fit, used to score every model in the λ sweep and to choose λ.
- **Test data**: a third set, used once, after λ has been chosen.

8.1.2 Model errors

All errors use Equation (25). When there are several variables (for Lorenz x, y, z), Rel is computed for each variable separately, with its own band, and the results are averaged.

- E_{train}: Rel of $\Theta(X)\Xi$ against the derivative it was fitted to, on the training data.
- E_{dv} (derivative on validation data): Rel of $\Theta(X)\Xi$ against the derivative of the validation data, with Θ evaluated at the true states of the validation data.
- E_{sim} (simulation on validation data): the SINDy equations are solved from the starting point of the validation data, and the solution is compared with the true states.

They are written out in full for Lorenz in Section 8.2.3 and for the PDE in Section 8.3.3.

8.1.3 The λ grid

λ is swept on a log scale with four values per decade, $\lambda_k = 10^{k/4}$, so each step multiplies or divides λ by $10^{0.25} \approx 1.78$:

$$\ldots,\ 0.178,\ 0.316,\ 0.562,\ 1.0,\ 1.78,\ 3.16,\ 5.62,\ 10,\ \ldots$$

8.1.4 Rule for choosing λ

1. **Sweep.** Run STLSQ for every λ on the grid.
2. **Merge.** λ values that give the same model are merged into one model with a λ range.
3. **Score.** Compute each model's error on the validation data (E_{dv} for Lorenz, E_{sim} for the PDE).

4. **Discard unstable models.** Remove models whose solution grows without bound when they are solved. The empty model (all coefficients zero) is kept.
5. **Order.** Arrange the remaining models from largest to smallest λ.
6. **Find the knee.** Between each pair of neighbours compute the drop in error, $\log_{10} E_{\text{before}} - \log_{10} E_{\text{after}}$, and find the largest drop.
7. **Select.** Take the model just after the largest drop.
8. **Choose λ.** Use the largest λ of that model's range.
9. **Test.** Score the selected model once on the test data.

This rule is called R4 below. Three other rules are compared with it for Lorenz in Section 8.2.5.

8.2 Check on the Lorenz system

8.2.1 System, data and derivatives

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = x(\rho - z) - y, \qquad \dot{z} = xy - \beta z, \qquad \sigma = 10,\ \rho = 28,\ \beta = 8/3, \tag{26}$$

starting from $x(0) = -8$, $y(0) = 7$, $z(0) = 27$. With these values the true equations are $\dot{x} = -10x + 10y$, $\dot{y} = 28x - y - xz$, $\dot{z} = -2.6667z + xy$: seven terms in total.

- Training data every $\Delta t = 0.001$ up to $t = 100$ (10^5 points). Library: all polynomials in x, y, z up to fifth order (56 terms, from 1 to z^5). 10 threshold-and-refit passes.

The derivatives $\dot{X}$ are obtained in three ways:

$$\begin{aligned}
\text{A (exact):} \quad & \dot{X} \text{ from the equations,} \\
\text{C2 (2nd-order central):} \quad & \dot{X}_i \approx \frac{X_{i+1} - X_{i-1}}{2\Delta t}, \qquad \text{error} \propto \Delta t^2, \\
\text{C4 (4th-order central):} \quad & \dot{X}_i \approx \frac{-X_{i+2} + 8X_{i+1} - 8X_{i-1} + X_{i-2}}{12\,\Delta t}, \qquad \text{error} \propto \Delta t^4.
\end{aligned}$$

For C2 and C4 the one or two points at each end, where the central formula cannot be used, are left out.

Equation	Term	True	A (exact)	C2 (2nd order)	C4 (4th order)
$\dot{x}$	x	-10.0000	-10.000000	-9.999780	-10.000000
$\dot{x}$	y	10.0000	10.000000	9.999779	10.000000
$\dot{y}$	x	28.0000	28.000000	27.997984	28.000000
$\dot{y}$	y	-1.0000	-1.000000	-0.999606	-1.000000
$\dot{y}$	xz	-1.0000	-1.000000	-0.999944	-1.000000
$\dot{z}$	z	-2.6667	-2.666667	-2.666589	-2.666667
$\dot{z}$	xy	1.0000	1.000000	0.999971	1.000000
$\max \lvert\xi_{\text{SINDy}} - \xi_{\text{true}}\rvert$			5.2×10^{-14}	2.0×10^{-3}	2.3×10^{-7}

Table 5: Lorenz coefficients of the model selected by R4 (Section 8.2.6). All other 161 coefficients are zero. Last row: maximum coefficient error (Section 8.2.3).

8.2.2 Training, validation and test runs

- **Training:** the run above, from $(-8, 7, 27)$, $0 \le t \le 100$.
- **Validation:** three runs from $(5, -5, 20)$, $(-3, 4, 30)$ and $(8, 8, 15)$, $0 \le t \le 20$, $\Delta t = 0.001$.
- **Test:** three runs from $(1, -1, 25)$, $(-6, 6, 18)$ and $(10, 3, 32)$, $0 \le t \le 20$, $\Delta t = 0.001$.

8.2.3 Error measures

$\dot{x}_{\text{ref}}$ is the derivative of the data, taken by the same method as in training (A, C2 or C4), and $\dot{x}_{\text{SINDy}} = \Theta(x, y, z)\,\xi_x$ is the SINDy equation for $\dot{x}$, evaluated at the true x, y, z of the data; likewise for y and z. The errors are given as fractions.

Training error, on the training run:

$$E_{train} = \frac{1}{3}\left[\frac{\text{RMS}(\dot{x}_{\text{SINDy}} - \dot{x}_{\text{ref}})}{\max \dot{x}_{\text{ref}} - \min \dot{x}_{\text{ref}}} + \frac{\text{RMS}(\dot{y}_{\text{SINDy}} - \dot{y}_{\text{ref}})}{\max \dot{y}_{\text{ref}} - \min \dot{y}_{\text{ref}}} + \frac{\text{RMS}(\dot{z}_{\text{SINDy}} - \dot{z}_{\text{ref}})}{\max \dot{z}_{\text{ref}} - \min \dot{z}_{\text{ref}}}\right].$$

Derivative error on validation data: the same formula on each validation run $j = 1, 2, 3$ (window: the whole run, $0 \le t \le 20$), averaged over the three runs:

$$E_{dv} = \frac{1}{3}\sum_{j=1}^{3} \frac{1}{3}\sum_{v\in\{x,y,z\}} \frac{\text{RMS}\left(\dot{v}_{\text{SINDy}} - \dot{v}_{\text{ref}}\right)_j}{\left(\max \dot{v}_{\text{ref}} - \min \dot{v}_{\text{ref}}\right)_j}.$$

Simulation error on validation data: the SINDy equations are solved from the starting point of validation run j, giving $x_{\text{SINDy}}(t), y_{\text{SINDy}}(t), z_{\text{SINDy}}(t)$, which are compared with the true $x(t), y(t), z(t)$ over $0 \le t \le T$, $T = 0.5$, 1 and 2:

$$E_{sim}(T) = \frac{1}{3}\sum_{j=1}^{3} \frac{1}{3}\sum_{v\in\{x,y,z\}} \frac{\text{RMS}_{[0,T]}\left(v_{\text{SINDy}} - v\right)_j}{\left(\max_{[0,T]} v - \min_{[0,T]} v\right)_j}.$$

A model whose solution grows without bound gets $E_{sim} = \infty$.

Maximum coefficient error: the largest of the $56 \times 3 = 168$ differences between the SINDy coefficients and the true ones,

$$\text{max coef. err.} = \max_{i,k} \left|\Xi_{ik,\text{SINDy}} - \Xi_{ik,\text{true}}\right|,$$

where the true Ξ has the 7 Lorenz coefficients and 161 zeros.

λ is chosen with E_{dv}; E_{sim} is reported alongside.

8.2.4 Sweeping the threshold

λ is swept from 10^{-12} to $10^{1.5} \approx 32$. Rows giving the same model are merged. "Wrong": a term not in the true equations. "Missing": a true term that was removed. Shaded: the model selected by R4 (Section 8.2.6).

λ range	terms	wrong	missing	max coef. err.	E_{train}	E_{dv}	$E_{sim}(0.5)$	$E_{sim}(1)$	$E_{sim}(2)$
1.0×10^{1} – 3.2×10^{1}	0	0	7	2.8×10^{1}	1.6×10^{-1}	1.6×10^{-1}	4.8×10^{-1}	3.9×10^{-1}	3.5×10^{-1}
3.2×10^{0} – 5.6×10^{0}	2	0	5	2.8×10^{1}	1.0×10^{-1}	1.0×10^{-1}	4.1×10^{-1}	3.3×10^{-1}	3.2×10^{-1}
1.8×10^{0}	3	0	4	3.1×10^{1}	1.0×10^{-1}	9.8×10^{-2}	4.5×10^{-1}	3.8×10^{-1}	3.1×10^{-1}
1.0×10^{0}	6	0	1	4.1×10^{1}	2.7×10^{-2}	2.8×10^{-2}	6.3×10^{-1}	6.1×10^{-1}	5.0×10^{-1}
1.0×10^{-12} – 5.6×10^{-1}	7	0	0	5.2×10^{-14}	2.2×10^{-16}	2.2×10^{-16}	4.8×10^{-11}	5.8×10^{-11}	2.0×10^{-10}

Table 6: Case A, exact derivative. Errors as fractions.

λ range	terms	wrong	missing	max coef. err.	E_{train}	E_{dv}	$E_{sim}(0.5)$	$E_{sim}(1)$	$E_{sim}(2)$
1.0×10^{1} – 3.2×10^{1}	0	0	7	2.8×10^{1}	1.6×10^{-1}	1.6×10^{-1}	4.8×10^{-1}	3.9×10^{-1}	3.5×10^{-1}
3.2×10^{0} – 5.6×10^{0}	2	0	5	2.8×10^{1}	1.0×10^{-1}	1.0×10^{-1}	4.1×10^{-1}	3.3×10^{-1}	3.2×10^{-1}
1.0×10^{0} – 1.8×10^{0}	3	0	4	3.1×10^{1}	1.0×10^{-1}	9.8×10^{-2}	4.5×10^{-1}	3.8×10^{-1}	3.1×10^{-1}
5.6×10^{-5} – 5.6×10^{-1}	7	0	0	2.0×10^{-3}	4.8×10^{-6}	4.9×10^{-6}	2.3×10^{-5}	5.4×10^{-5}	1.0×10^{-4}
3.2×10^{-5}	8	1	0	2.0×10^{-3}	4.6×10^{-6}	4.7×10^{-6}	2.1×10^{-5}	6.1×10^{-5}	1.4×10^{-4}
1.8×10^{-5}	9	2	0	2.0×10^{-3}	4.4×10^{-6}	4.5×10^{-6}	2.1×10^{-5}	5.4×10^{-5}	1.3×10^{-4}
1.0×10^{-5}	12	5	0	2.8×10^{-3}	3.8×10^{-6}	3.9×10^{-6}	4.6×10^{-5}	7.1×10^{-5}	1.7×10^{-4}
5.6×10^{-6}	14	7	0	2.8×10^{-3}	2.3×10^{-6}	2.5×10^{-6}	4.7×10^{-5}	5.7×10^{-5}	1.3×10^{-4}
1.8×10^{-6} – 3.2×10^{-6}	17	10	0	1.2×10^{-3}	1.2×10^{-6}	1.4×10^{-6}	1.8×10^{-5}	4.5×10^{-5}	8.6×10^{-5}
1.8×10^{-7} – 1.0×10^{-6}	19	12	0	3.7×10^{-3}	3.3×10^{-7}	3.6×10^{-7}	1.8×10^{-5}	2.7×10^{-5}	3.1×10^{-5}
1.8×10^{-9} – 1.0×10^{-7}	21	14	0	1.8×10^{-3}	1.3×10^{-10}	1.4×10^{-10}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
1.0×10^{-9}	22	15	0	1.8×10^{-3}	1.3×10^{-10}	1.3×10^{-10}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
5.6×10^{-10}	23	16	0	1.8×10^{-3}	1.3×10^{-10}	1.3×10^{-10}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
3.2×10^{-10}	24	17	0	1.8×10^{-3}	1.3×10^{-10}	1.3×10^{-10}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
1.0×10^{-10}	32	25	0	1.8×10^{-3}	1.1×10^{-10}	1.2×10^{-10}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
5.6×10^{-11}	42	35	0	1.8×10^{-3}	1.0×10^{-10}	1.1×10^{-10}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
3.2×10^{-11}	38	31	0	1.8×10^{-3}	8.7×10^{-11}	9.8×10^{-11}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
1.8×10^{-11}	46	39	0	1.8×10^{-3}	3.2×10^{-11}	3.9×10^{-11}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
1.0×10^{-11}	53	46	0	1.8×10^{-3}	3.0×10^{-11}	3.7×10^{-11}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
5.6×10^{-12}	60	53	0	1.8×10^{-3}	2.1×10^{-11}	2.6×10^{-11}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
3.2×10^{-12}	62	55	0	1.8×10^{-3}	1.6×10^{-11}	2.0×10^{-11}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
1.8×10^{-12}	58	51	0	1.8×10^{-3}	6.5×10^{-12}	9.4×10^{-12}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}
1.0×10^{-12}	64	57	0	1.8×10^{-3}	6.4×10^{-12}	9.5×10^{-12}	2.3×10^{-5}	3.1×10^{-5}	3.5×10^{-5}

Table 7: Case C2, 2nd-order finite differences.

λ range	terms	wrong	missing	max coef. err.	E_{train}	E_{dv}	$E_{sim}(0.5)$	$E_{sim}(1)$	$E_{sim}(2)$
1.0×10^{1} – 3.2×10^{1}	0	0	7	2.8×10^{1}	1.6×10^{-1}	1.6×10^{-1}	4.8×10^{-1}	3.9×10^{-1}	3.5×10^{-1}
3.2×10^{0} – 5.6×10^{0}	2	0	5	2.8×10^{1}	1.0×10^{-1}	1.0×10^{-1}	4.1×10^{-1}	3.3×10^{-1}	3.2×10^{-1}
1.0×10^{0} – 1.8×10^{0}	3	0	4	3.1×10^{1}	1.0×10^{-1}	9.8×10^{-2}	4.5×10^{-1}	3.8×10^{-1}	3.1×10^{-1}
1.8×10^{-8} – 5.6×10^{-1}	7	0	0	2.3×10^{-7}	1.1×10^{-9}	1.1×10^{-9}	2.7×10^{-9}	5.7×10^{-9}	1.2×10^{-8}
1.0×10^{-8}	9	2	0	8.5×10^{-7}	1.1×10^{-9}	1.1×10^{-9}	2.5×10^{-9}	5.6×10^{-9}	1.7×10^{-8}
5.6×10^{-9}	10	3	0	8.5×10^{-7}	1.1×10^{-9}	1.1×10^{-9}	2.5×10^{-9}	5.5×10^{-9}	1.7×10^{-8}
3.2×10^{-9}	13	6	0	8.5×10^{-7}	9.7×10^{-10}	1.0×10^{-9}	5.1×10^{-9}	6.4×10^{-9}	1.8×10^{-8}
1.8×10^{-9}	18	11	0	8.8×10^{-7}	5.5×10^{-10}	5.9×10^{-10}	4.9×10^{-9}	7.4×10^{-9}	1.2×10^{-8}
1.0×10^{-9}	19	12	0	1.4×10^{-6}	5.5×10^{-10}	5.9×10^{-10}	4.8×10^{-9}	6.9×10^{-9}	1.0×10^{-8}
5.6×10^{-10}	22	15	0	1.4×10^{-6}	6.0×10^{-10}	6.4×10^{-10}	3.7×10^{-9}	6.0×10^{-9}	1.4×10^{-8}
3.2×10^{-10}	32	25	0	9.0×10^{-7}	4.7×10^{-10}	4.9×10^{-10}	2.9×10^{-9}	4.2×10^{-9}	1.5×10^{-8}
1.8×10^{-10}	38	31	0	5.6×10^{-7}	3.6×10^{-10}	3.8×10^{-10}	8.1×10^{-9}	1.4×10^{-8}	1.9×10^{-8}
1.0×10^{-10}	46	39	0	9.8×10^{-8}	2.3×10^{-10}	2.5×10^{-10}	4.2×10^{-9}	5.5×10^{-9}	9.2×10^{-9}
5.6×10^{-11}	51	44	0	7.9×10^{-7}	1.3×10^{-10}	1.6×10^{-10}	2.2×10^{-9}	3.7×10^{-9}	7.3×10^{-9}
3.2×10^{-11}	56	49	0	7.4×10^{-7}	1.2×10^{-10}	1.5×10^{-10}	2.1×10^{-9}	2.6×10^{-9}	4.0×10^{-9}
1.8×10^{-11}	60	53	0	7.7×10^{-7}	8.4×10^{-11}	1.0×10^{-10}	3.1×10^{-9}	3.3×10^{-9}	5.5×10^{-9}
1.0×10^{-11}	62	55	0	2.1×10^{-6}	3.9×10^{-11}	5.4×10^{-11}	1.8×10^{-9}	2.7×10^{-9}	4.7×10^{-9}
5.6×10^{-12}	64	57	0	2.9×10^{-6}	2.4×10^{-11}	3.6×10^{-11}	1.4×10^{-9}	2.0×10^{-9}	4.0×10^{-9}
3.2×10^{-12}	53	46	0	1.2×10^{-7}	6.8×10^{-12}	9.4×10^{-12}	2.6×10^{-9}	2.9×10^{-9}	6.1×10^{-9}
1.8×10^{-12}	58	51	0	1.3×10^{-7}	5.8×10^{-12}	7.7×10^{-12}	2.3×10^{-9}	2.4×10^{-9}	5.0×10^{-9}
1.0×10^{-12}	58	51	0	1.3×10^{-7}	5.8×10^{-12}	7.7×10^{-12}	2.3×10^{-9}	2.4×10^{-9}	5.0×10^{-9}

Table 8: Case C4, 4th-order finite differences.

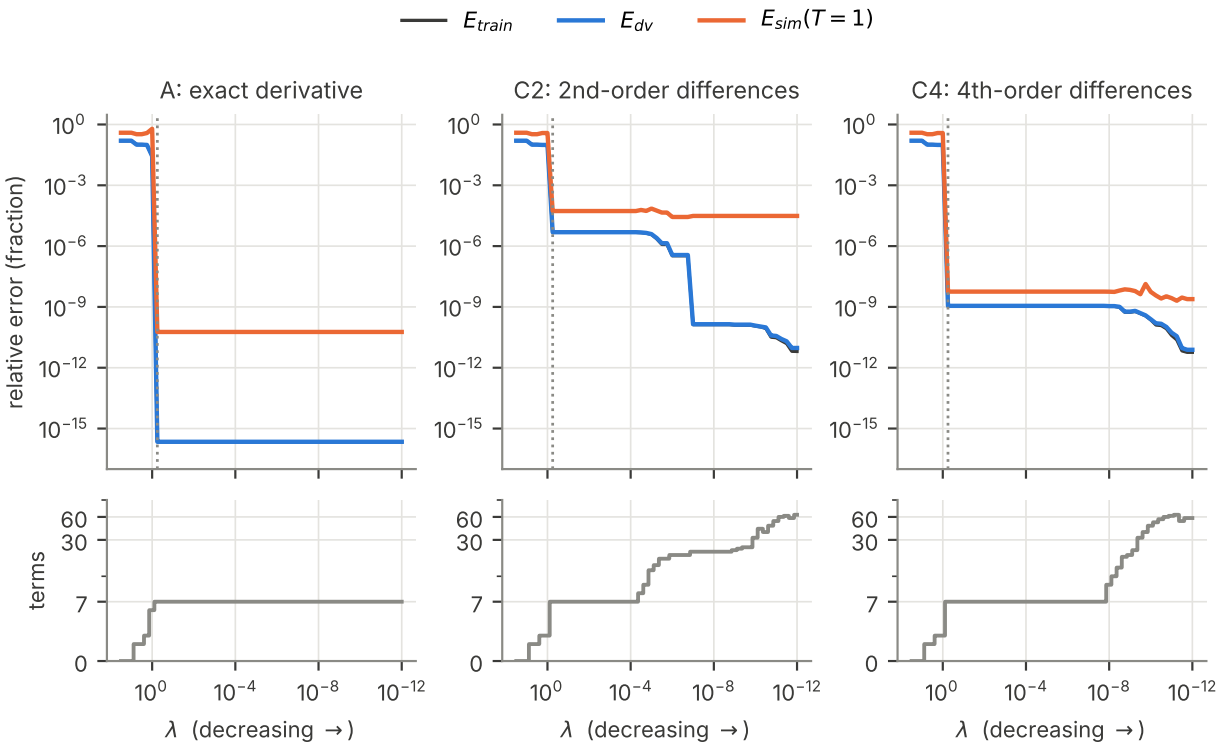


Figure 37: The three error measures against λ (λ decreases to the right). Lower panels: number of terms. Dotted line: $\lambda = 0.56$.

8.2.5 Comparison of rules for choosing λ

Four rules, each applied to E_{dv} and to $E_{sim}(T{=}1)$:

1. **R1:** the largest λ whose error is within 10% of the smallest error.
2. **R2:** the largest λ whose error is within a factor 10 of the smallest error.
3. **R3:** the largest λ whose error is below a fixed level, 10^{-3}.
4. **R4:** the model just after the largest drop in error (Section 8.1.4).

Case	Error	R1	R2	R3	R4
A	E_{dv}	7 (0)	7 (0)	7 (0)	7 (0)
A	$E_{sim}(T{=}1)$	7 (0)	7 (0)	7 (0)	7 (0)
C2	E_{dv}	58 (51)	46 (39)	7 (0)	7 (0)
C2	$E_{sim}(T{=}1)$	19 (12)	7 (0)	7 (0)	7 (0)
C4	E_{dv}	58 (51)	62 (55)	7 (0)	7 (0)
C4	$E_{sim}(T{=}1)$	64 (57)	7 (0)	7 (0)	7 (0)

Table 9: Number of terms (number of wrong terms) of the model picked by each rule. The true model has 7 terms.

8.2.6 Applying R4 to Lorenz

Case	Score	terms	wrong	λ range of the selected model	selected λ
A	E_{dv}	7	0	1.0×10^{-12} – 5.6×10^{-1}	5.62×10^{-1}
A	$E_{sim}(T{=}1)$	7	0	1.0×10^{-12} – 5.6×10^{-1}	5.62×10^{-1}
C2	E_{dv}	7	0	5.6×10^{-5} – 5.6×10^{-1}	5.62×10^{-1}
C2	$E_{sim}(T{=}1)$	7	0	5.6×10^{-5} – 5.6×10^{-1}	5.62×10^{-1}
C4	E_{dv}	7	0	1.8×10^{-8} – 5.6×10^{-1}	5.62×10^{-1}
C4	$E_{sim}(T{=}1)$	7	0	1.8×10^{-8} – 5.6×10^{-1}	5.62×10^{-1}

Table 10: R4 applied to every Lorenz case.

The coefficients of the selected models are given in Table 5.

8.2.7 Test runs

The selected model is scored once on the three test runs, with the same formulas as E_{dv} and E_{sim}. The true Lorenz equations are scored on the same runs in the same way.

Case	Model	E_{dv}	$E_{sim}(0.5)$	$E_{sim}(1)$	$E_{sim}(2)$
A	SINDy, selected (7 terms)	2.2×10^{-16}	4.7×10^{-11}	3.6×10^{-11}	9.4×10^{-11}
A	true Lorenz equations	2.0×10^{-17}	4.7×10^{-11}	3.6×10^{-11}	9.4×10^{-11}
C2	SINDy, selected (7 terms)	4.9×10^{-6}	3.7×10^{-5}	4.5×10^{-5}	1.4×10^{-4}
C2	true Lorenz equations	6.8×10^{-6}	4.7×10^{-11}	3.6×10^{-11}	9.4×10^{-11}
C4	SINDy, selected (7 terms)	1.1×10^{-9}	4.7×10^{-9}	4.8×10^{-9}	1.6×10^{-8}
C4	true Lorenz equations	1.2×10^{-9}	4.7×10^{-11}	3.6×10^{-11}	9.4×10^{-11}

Table 11: Test errors (fractions), mean over the three test runs. ODE solver tolerance: 10^{-10}.

8.3 SINDy on the PDE, one frequency at a time

SINDy is fitted separately at each of four frequencies, and the coefficients are compared with those of the Galerkin model.

8.3.1 Training runs

ω	period P	$\Delta\tau = P/200$	$\tau_{\max} = 5P$	points
0.1	62.83	0.3142	314.2	1001
1	6.283	0.03142	31.42	1001
10	0.6283	0.003142	3.142	1001
100	0.06283	0.0003142	0.3142	1001

Table 12: Training runs: $\delta = A\sin(\omega\tau)$, $f = 1$.

- Library $\{C_s,\ C_{\text{avg}},\ \delta\}$, the terms of the Galerkin model. SINDy fits only $dC_s/d\tau$; C_{avg} follows the mass balance, Equation (7).
- $dC_s/d\tau$ by 2nd-order differences at all 1001 points: central differences inside, one-sided 2nd-order differences at the first and last point.
- The PDE data for this section come from a Python solver: Chebyshev collocation in r with an implicit time integrator, tolerance 10^{-10}. Doubling the number of radial points changes C_s by less than 10^{-9}.

8.3.2 Validation and test data

Four validation runs at the same ω, each 10 periods long with $\Delta\tau = P/200$:

1. **cos**: $\delta = A\cos(\omega\tau)$, from a uniform particle.
2. **sin late**: periods 6 to 15 of the training sine (the PDE is continued from its state at $\tau = 5P$, and the model starts from the PDE's C_s and C_{avg} there).
3. **sin ph**: $\delta = A\sin(\omega\tau + \pi/4)$.
4. **cos ph**: $\delta = A\cos(\omega\tau + \pi/4)$.

"Stacked" is the mean of the four errors.

Test: the random multisine $\delta = s\sum_{k=1}^{3} a_k \sin(\omega_k\tau + \phi_k)$ with $\omega_k = \omega/2,\ \omega,\ 2\omega$, a_k and ϕ_k as in Table 13, scaled so that $\max|\delta| = A$, 10 periods of ω.

ω	ω_k	a_k	ϕ_k
0.1	0.05, 0.1, 0.2	0.82, 0.74, 0.75	4.62, 3.09, 0.48
1	0.5, 1, 2	0.54, 0.91, 0.82	1.24, 3.70, 5.63
10	5, 10, 20	0.91, 0.53, 0.69	4.99, 2.70, 4.75
100	50, 100, 200	0.62, 0.97, 0.53	1.31, 0.81, 2.65

Table 13: Parameters of the random (test) inputs.

8.3.3 Error measures

Equation (25) with a single variable, in %.

- E_{sim}: the SINDy equation for C_s is solved together with $dC_{\text{avg}}/d\tau = -3\delta$, from the PDE's C_s and C_{avg} at the start of the run, and compared with the PDE's C_s over the whole run:

$$E_{sim}\,(\%) = 100 \times \frac{\text{RMS}\left(C_{s,\text{SINDy}} - C_{s,\text{PDE}}\right)}{\max C_{s,\text{PDE}} - \min C_{s,\text{PDE}}} \quad \text{(whole run)}.$$

- E_{dv}: the SINDy right-hand side, evaluated at the PDE's C_s, C_{avg} and δ, against $dC_s/d\tau$ of the PDE data (2nd-order differences, as in training), whole run:

$$E_{dv}\,(\%) = 100 \times \frac{\text{RMS}\left(\xi_1 C_s + \xi_2 C_{\text{avg}} + \xi_3\delta - dC_s/d\tau\right)}{\max dC_s/d\tau - \min dC_s/d\tau}.$$

- E_{train}: the same as E_{dv} on the training run.

λ is chosen with the stacked E_{sim}.

8.3.4 Models found

		coefficients				E_{sim} (%) on validation runs					derivative error (%)	
ω	λ range	C_s	C_{avg}	δ	stable	cos	sin late	sin ph	cos ph	stacked	E_{dv}, stacked	E_{train}
0.1	1.8×10^{1} – 1.0×10^{3}	+0.000	+0.000	+0.000	yes	35	61	50	50	49	34.2	35.3
0.1	1.0×10^{1}	+14.998	-14.999	+0.000	no	∞	∞	∞	∞	∞	0.961	0.144
0.1	1.0×10^{-6} – 5.6×10^{0}	-34.638	+34.638	-9.927	yes	2.3×10^{-5}	1.7×10^{-5}	1.9×10^{-5}	2.7×10^{-5}	2.1×10^{-5}	1.26	9.06×10^{-3}
1	1.8×10^{1} – 1.0×10^{3}	+0.000	+0.000	+0.000	yes	35	61	50	50	49	24.4	35.3
1	1.0×10^{-6} – 1.0×10^{1}	-35.039	+35.039	-10.004	yes	5.4×10^{-3}	1.8×10^{-4}	3.8×10^{-3}	3.9×10^{-3}	3.3×10^{-3}	0.136	0.0436
10	1.8×10^{1} – 1.0×10^{3}	+0.000	+0.000	+0.000	yes	35	57	44	45	45	17.4	35.2
10	1.0×10^{-6} – 1.0×10^{1}	-38.077	+38.077	-10.313	yes	0.25	0.042	0.19	0.19	0.17	0.999	0.641
100	3.2×10^{1} – 1.0×10^{3}	+0.000	+0.000	+0.000	yes	35	52	35	35	39	16.8	34.5
100	1.8×10^{1}	+28.745	-28.738	+0.000	no	∞	∞	∞	∞	∞	16.4	33.5
100	1.0×10^{-6} – 1.0×10^{1}	-133.216	+133.207	-17.630	yes	1.7	1.4	1.5	2.1	1.7	1.86	2.82

Table 14: Models found in the λ sweep (10^{-6} to 10^{3}) and their errors in %. Each validation column is that model's E_{sim} on that run. ∞: the solution grows without bound. The empty model keeps C_s at its starting value. Shaded: the model selected by R4.

8.3.5 Choosing λ

ω	E_{sim} on cos	sin late	sin ph	cos ph	**stacked**	E_{dv}, stacked step 4 skipped	step 4 used	λ range	selected λ
0.1	3	3	3	3	3	2, **unstable**	3	1.0×10^{-6} – 5.6×10^{0}	5.6×10^{0}
1	3	3	3	3	3	3	3	1.0×10^{-6} – 1.0×10^{1}	1.0×10^{1}
10	3	3	3	3	3	3	3	1.0×10^{-6} – 1.0×10^{1}	1.0×10^{1}
100	3	3	3	3	3	3	3	1.0×10^{-6} – 1.0×10^{1}	1.0×10^{1}

Table 15: Number of terms of the model selected by R4 for each choice of validation error, with and without step 4 (removal of unstable models). The selection used is E_{sim}, stacked.

8.3.6 Test on the random input

The selected model at each ω is scored once on the random multisine; the Galerkin model is scored on the same runs.

ω	E_{sim} (%) Galerkin	SINDy	E_{dv} (%) Galerkin	SINDy	E_{sim}: Galerkin / SINDy
0.1	2.02×10^{-6}	1.81×10^{-5}	0.583	0.576	0.11
1	2.18×10^{-3}	1.79×10^{-3}	0.143	0.13	1.2
10	0.756	0.563	1.61	1.46	1.3
100	11.5	3.93	4.03	2.38	2.9

Table 16: Test errors on the random multisine (10 periods, whole run), in %.

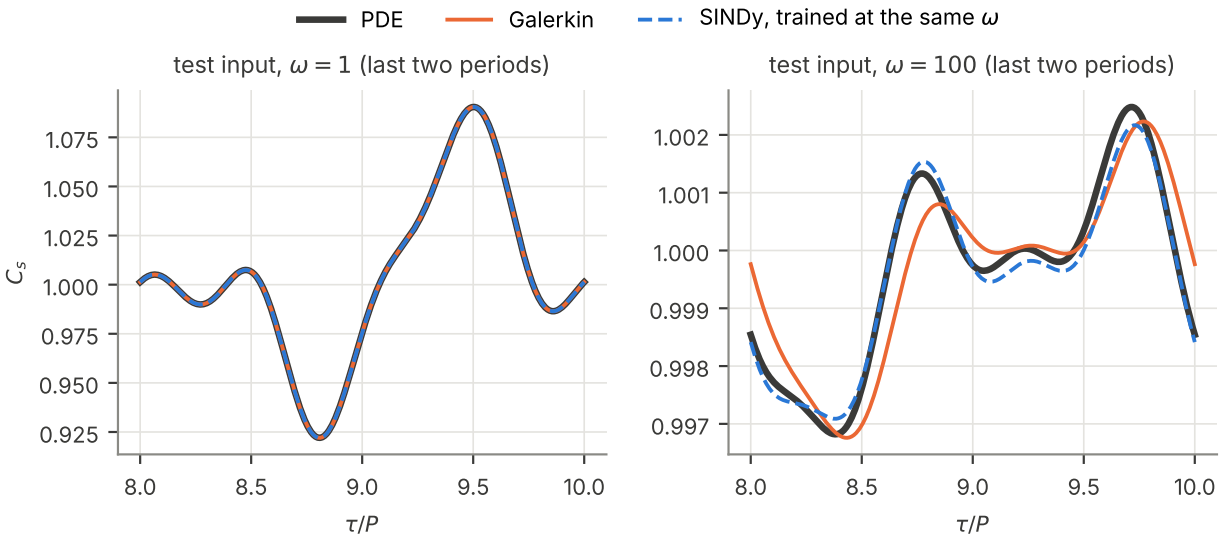


Figure 38: C_s on the test input over its last two periods, at $\omega = 1$ and $\omega = 100$.

8.3.7 Selected equations and the Galerkin model

	derivative	C_s	C_{avg}	δ
Galerkin	–	−35	+35	−10
$\omega = 0.1$	2nd order	-34.638	+34.638	-9.927
	4th order	-34.665	+34.665	-9.933
	2nd order, end points dropped	-34.996	+34.996	-9.999
$\omega = 1$	2nd order	-35.039	+35.039	-10.004
	4th order	-35.048	+35.048	-10.006
	2nd order, end points dropped	-35.040	+35.040	-10.004
$\omega = 10$	2nd order	-38.077	+38.077	-10.313
	4th order	-38.084	+38.084	-10.315
	2nd order, end points dropped	-38.078	+38.078	-10.313
$\omega = 100$	2nd order	-133.216	+133.207	-17.630
	4th order	-133.239	+133.229	-17.633
	2nd order, end points dropped	-133.225	+133.215	-17.630

Table 17: Coefficients of the selected model at each training frequency. "End points dropped": the same 2nd-order fit without the first and last point.

8.3.8 Cross-test: trained at one frequency, tested at another

Every selected model is tested on the cosine run at each ω, over the last period, with two error measures:

- C_s: Rel (%) of C_s, Equation (25), last period.
- gap: the same RMS error divided by the band of the gap $C_s - C_{\text{avg}}$ instead of the band of C_s.

Model	test $\omega = 0.1$		test $\omega = 1$		test $\omega = 10$		test $\omega = 100$	
	C_s	gap	C_s	gap	C_s	gap	C_s	gap
Galerkin	1.52×10^{-6}	1.61×10^{-4}	1.07×10^{-3}	0.0161	0.704	1.48	16.3	20.1
SINDy, trained at $\omega = 0.1$	2.18×10^{-5}	3.17×10^{-3}	1.15×10^{-3}	0.0174	0.715	1.51	16.4	20.3
SINDy, trained at $\omega = 1$	1.23×10^{-4}	0.0184	1.85×10^{-4}	2.72×10^{-3}	0.695	1.46	16.3	20.1
SINDy, trained at $\omega = 10$	9.40×10^{-3}	1.41	0.0925	1.39	0.0423	0.089	15.6	19.3
SINDy, trained at $\omega = 100$	0.107	16	1.06	16	7.29	15.4	1.43	1.77

Table 18: Cross-test, Rel (%), cosine flux, last period.

9 Summary

- Changing the flux shape changes the surface concentration response. The average concentration is set by the total charge applied, so matched pairs with the same total charge end at the same C_{avg} but different C_s.
- Increasing the frequency of an oscillating flux reduces the visible oscillation in C_s and confines the response to a thinner layer near the surface, with the material model unchanged.
- Increasing the constant diffusivity makes the radial profile more uniform. The maximum gap $C_{\text{avg}} - C_s$ falls from 0.0710 at $f = 0.01$ to 0.000051 at $f = 100$.
- The average concentration follows the applied flux and is essentially independent of the diffusivity, as Equation (7) requires.
- `trueRandom` gives different reproducible excitation signals for different seeds, with surface responses that are similar in character but different in detail.

- The Galerkin method with the two shapes 1 and $r^2 - 3/5$ turns the PDE into two ODEs: $dC_{\mathrm{avg}}/d\tau = -3\delta$ and $dC_s/d\tau = -35 f(C_s - C_{\mathrm{avg}}) - 10\,\delta$.
- SINDy with the library $\{C_s, C_{\mathrm{avg}}, \delta\}$ was fitted on the Lorenz system and on PDE data at $\omega = 0.1$, 1, 10 and 100, with λ chosen by rule R4; the results are in Tables 5 and 17.

A Code for Experiment 6

- **Python**: `pde.py` (PDE solver and Galerkin model), `common.py` (error formula, STLSQ, rules R1–R4), `lorenz.py`, `pde_sindy.py`, `figures.py`, `tables.py`.
- **MATLAB**: `relErr.m`, `stlsq.m`, `lambdaGrid.m`, `sweepLambda.m`, `selectLambdaR4.m`, `solvePDE.m` (uses `pdepe` when available, otherwise a Chebyshev method of lines), `galerkinModel.m`, `run_lorenz.m`, `run_pde_omega.m`, `run_pde_all.m`, `test_quick.m`.